

The ESM-2 model series

Comparison between ESM-2 and ESM-1b

Feature	ESM-1b	ESM-2	What Changed
Architecture	BERT-style encoder-only Transformer	BERT-style encoder-only Transformer	-
Positional Encoding	Learned positional embeddings	Rotary Positional Embeddings (RoPE)	RoPE improves extrapolation beyond max. sequence length
Dropout	Word-level dropout for masked tokens	Removed entirely	Lower computational cost without degrading performance
Tied input/output embeddings	Yes	Yes	-
Optimization	Adam with warmup + inverse sqrt decay	Adam with warmup + linear decay	-
Training Duration	56 epochs (~500K steps)	500K steps (270K for 15B model)	ESM-2 uses larger batch sizes (131K vs. 2M tokens)
Training Dataset	~30M sequences	~60M sequences	Increased sequence diversity
Model Scaling	Up to 650M parameters	8M → 15B parameter models	A variety of models with different sizes (including much larger)

<u>Model</u>	<u>Parameters</u>	<u>Layers</u>	<u>Hidden Dim</u>	<u>Attention Heads</u>
ESM-2-8M	8M	6	320	20
ESM-2-35M	35M	12	480	20
ESM-2-150M	150M	30	640	20
ESM-2-650M	650M	33	1280	20
ESM-2-3B	3B	36	2560	40
ESM-2-15B	15B	48	5120	40

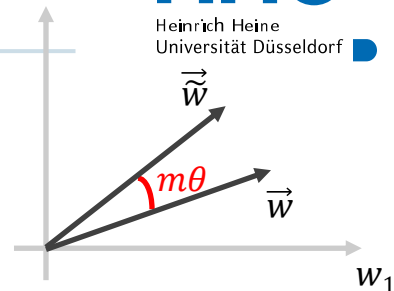
RoPE positional encoding

- Rotary Positional Embeddings (RoPE)

m position in input sequence

θ rotation constant

$$\vec{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \xrightarrow{\text{rotate by } m \cdot \theta} \vec{\tilde{w}}$$



- The rotations are applied to pairs of two values

- $[x_1, x_2, x_3, x_4, \dots, x_{d-1}, x_d] \rightarrow [(x_1, x_2), (x_3, x_4), \dots, (x_{d-1}, x_d)]$

- Each pair (x_{2k}, x_{2k+1}) is rotated by an angle $\theta_k = m \cdot \frac{1}{10000^{2k/d}}$

- Low dimensions rotate slowly $\rightarrow$ capture long-range patterns

- High dimensions rotate quickly $\rightarrow$ capture fine-grained local positions.

- RoPE embeddings are applied in each encoder layer before applying attention

- Applied to queries and keys right before computing the attention scores

- Not applied to the values V (the token embeddings) directly

$$Attention(Q, K, V) = softmax(\frac{Q^T \cdot K}{\sqrt{d}}) \cdot V^T$$

RoPE (2)

- Advantages of RoPE
 - Rotation naturally captures relative position — how far apart two tokens are.
 - It works well with attention because attention cares about token-to-token relations, not absolute positions.
 - RoPE is also generalizable and can work with sequences longer than training.
- Sequence truncation in ESM-2:
 - Proteins truncated during training to 1024 tokens
 - After training: RoPE allows for extrapolation to longer sequences beyond the training context window

<u>Concept</u>	<u>RoPE Behavior</u>
Applied to token embeddings directly	No (no modified token representation)
Applied to values (V)	No
Applied to Q and K	Yes
When applied	Every layer before attention dot-product
Encodes absolute or relative position?	Relative
Affects attention score computation	Yes

Other Protein Language Models

- ProtT5 (ProtT5-XL-U50)
 - Training task:
 - Masked language modeling with masking probability of 15%
 - Training data:
 - UniRef50: ~45M unique protein sequences
 - Model architecture:
 - T5 Transformer Network consisting of encoder and decoder (for training)
 - Only uses the encoder for generating protein representations
 - 3B learnable parameters

Comparison between protein language models

Model	# Params	# Updates	validation Perplexity	LR P@L	LR P@L/5	CASP14	CAMEO
ESM-2	8M	500K	10.33	0.17	0.29	0.37	0.48
	35M	500K	8.95	0.30	0.51	0.41	0.56
	150M	500K	7.75	0.44	0.70	0.49	0.65
	650M	500K	6.95	0.52	0.79	0.51	0.70
	3B	500K	6.49	0.54	0.81	0.52	0.72
	15B	270K	6.37	0.54	0.82	0.55	0.72
ESM-1b	650M	—	—	0.41	0.66	0.42	0.64
Prot-T5-XL (UR50) (21)	3B	—	—	0.48	0.72	0.50	0.69
Prot-T5-XL (BFD) (21)	3B	—	—	0.36	0.58	0.46	0.63
CARP (24)	640M	—	—	—	—	0.42	0.59